

Contact

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Top Skills

Blockchain
API Development
Prometheus

Sri Hari Sirisipalli

AI Systems Engineer | ML Systems & LLM Infrastructure | Scalable Experimentation & Retrieval Architectures
United States

Summary

I build AI systems that move beyond models into structured, reproducible infrastructure.

Over the past 4 years, I've designed and implemented large-scale ML pipelines spanning surrogate modeling, retrieval-based LLM systems, and cloud-native AI platforms. My work centers on disciplined experimentation — executing thousands of controlled model runs, applying feature reduction at scale, and building retrain-to-predict architectures with strict validation isolation.

I have built offline RAG systems integrating embedding workflows, FAISS-based vector indexing, and secure inference stacks, as well as ML pipelines involving 3,500+ structured experiments and over 40,000 tracked training runs. I focus on modular system design, reproducibility, and balancing model performance with operational reliability.

Technically, my experience spans ML systems engineering, LLM retrieval architectures, AWS infrastructure automation, experiment orchestration, and engineering automation workflows.

I am particularly interested in ML systems and LLM infrastructure roles where experimentation rigor, system reliability, and scalable design intersect.

Experience

Pangeon

Software Engineer – GenAI & Cloud Infrastructure
March 2024 - Present (2 years)

Built LLM-driven retrieval systems and scalable cloud infrastructure supporting semantic similarity and knowledge discovery workflows.

- Designed patent similarity platform integrating embedding pipelines with FAISS-based vector indexing for large-scale semantic retrieval.
- Developed embedding generation and storage workflows supporting scalable similarity search across domain corpora.
- Reduced AWS infrastructure costs by >30% through EC2 autoscaling and compute right-sizing strategies.
- Implemented Terraform-based infrastructure-as-code for reproducible environment provisioning and IAM-controlled deployments.
- Evaluated serverless vector database architectures for intermittent batch-oriented LLM workloads, analyzing cold-start latency and cost trade-offs.

VVS Systems

AI Systems Engineer – Offshore Digital Twin Systems

June 2025 - February 2026 (9 months)

Designed and implemented scalable surrogate ML systems for high-variance simulation modeling within digital twin environments. Led large-scale experimentation workflows to optimize predictive performance under strict validation controls.

- Executed 3,525 controlled ML→DL experiments, optimizing a multi-target MLP achieving $R^2 = 0.8086$.
- Built time-series training framework over 3,888 simulation files (4 Hz sampling), conducting 245 structured model sweeps and deploying a Random Forest achieving $R^2 = 0.9992$ (RMSE = 0.4383 ft).
- Engineered 156-feature space across spectral, kinematic, and signal domains; performed 166-round Recursive Feature Elimination (~1.08M model fits) to reduce to 22 production-grade features.
- Developed hierarchical classification–regression architecture (171 RFE rounds, 36,752 experiments) achieving 100% test accuracy and 41% reduction in worst-case angular error.
- Built modular retrain-to-predict ML infrastructure with artifact versioning, CLI-driven execution, and reproducible tracking across 40K+ experiment runs.

Applied AI Research & Engineering

AI Systems Engineer

June 2023 - June 2025 (2 years 1 month)

Designed applied AI systems spanning predictive maintenance, simulation automation, and offline LLM-based conversational interfaces.

- Built anomaly detection pipelines using accelerometer sensor data with RMS, FFT, kurtosis, and statistical feature engineering for condition monitoring.

- Trained and benchmarked Random Forest, SVM, k-NN, and deep learning models with structured cross-validation and expert-in-the-loop validation workflows.
- Automated engineering simulation pipelines using MOSES, ANSYS APDL, and FreeCAD, enabling parametric design-to-analysis execution.
- Developed offline 3D AI avatar integrating Whisper (STT), Ollama-hosted LLM, LangChain RAG, and Silero TTS, reducing STT latency by 75% and TTS latency by 52%.
- Designed RAG control-gate mechanisms to enforce validated knowledge retrieval in secure offline environments.

SAS2PY

Data Engineer

March 2023 - June 2023 (4 months)

Hyderabad, Telangana, India

- Efficient SAS-to-PySpark Migration Translated complex SAS scripts into optimized PySpark code, improving scalability and performance for large-scale data processing.
- Automated Validation Scripts: Developed comprehensive scripts to compare SAS and PySpark outputs with high precision, generating HTML reports that highlight discrepancies to ensure accuracy and validate migration results.
- Data Pipeline Dependency Mapping: Applied graph theory to uncover and visualize intricate file dependencies within data pipelines, enabling bottleneck identification and enhancing overall efficiency.

Corteva Agriscience

Machine Learning Intern

July 2022 - December 2022 (6 months)

Hyderabad, Telangana, India

- Cross-Framework Model Conversion: Converted deep learning models into multiple formats, facilitating seamless deployment across diverse platforms and frameworks.
- Framework-Agnostic Model Integration: Ensured smooth transitions of models from TensorFlow and PyTorch to various deployment environments, optimizing cross-platform compatibility.
- Computer Vision Model Optimization: Enhanced the performance of computer vision models by applying advanced techniques such as model compression, quantization, and pruning.

- Efficiency without Compromise: Improved model efficiency while maintaining high accuracy, ensuring robust and reliable deployment in resource-constrained environments.

Dojima Network

Software Engineer Internship

June 2022 - December 2022 (7 months)

Bangalore Urban, Karnataka, India

- Polkadot-Dojima Integration: Seamlessly integrated the Polkadot ecosystem with the Dojima network chain, enabling enhanced interoperability and functionality.
- API Monitoring Dashboards: Designed and implemented Prometheus-Grafana dashboards to monitor APIs in real-time, improving system observability and performance tracking

Education

Mahindra University

Bachelor's degree, Mechanical Engineering · (June 2018 - July 2022)

Sri Chaitanya College of Education

Intermediate, MPC · (April 2015 - May 2017)